

BASF

We create chemistry

Floragenist® Blossom

The Skin Guardian Elixir

Global Except US

TABLE OF CONTENTS

Summary file 3

Floragenist range - Next-generation botanical extracts reinvented 4

Floragenist Blossom - Orange Blossom perfected by science 5

AquaGenesis technology: use of coconut water as natural solvent for enriched plant extracts 10

Demonstrated efficacy 14

Epidermal thickness increase (*in vitro*) 15

Reduction of skin permeability induced by SLS stress (*in vitro*) 16

Filaggrin recovery after mechanical stress (*in vitro*) 17

Filaggrin protection from dysbiosis (*in vitro*) 19

Stimulation of Corneodesmosin protein on keratinocytes (*in vitro*) 20

Stimulation of lncRNA H19, Desmoglein 1 and Involucrin on keratinocytes (*in vitro*) 21

Stimulation of 12 genes implicated in cohesion and differentiation process (*in vitro*) 23

Barrier strengthening (*in vivo*) 25

General conclusion 27

Annexes 28

Bibliography 30

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SUMMARY FILE

Floragenist Blossom	BC10191
Origin - Description	Natural *Citrus aurantium* amara bud extract produced by AquaGenesis patented extraction technology enriched in Flavanone glycosides
Regulatory data	
INCI	Maltodextrin (and) Cocos Nucifera (Coconut) Fruit Extract (and) Citrus Aurantium Amara (Bitter Orange) Flower Extract
China	All components are listed in Inventory of Existing Cosmetic Ingredient in China (IECIC)
CAS# EINECS	9050-36-6; 8001-31-8; 72968-50-4 232-940-4; 232-282-8; 277-143-2
Appearance	Beige to brown powder with characteristic odor
Natural labels	Raw material conforms to Cosmos standards of natural and organic cosmetics
Naturalness content (ISO 16128)	100% from natural origin
Cosmetic use	
Properties	• Helps to protect epidermal keratinocytes against *S. aureus* • Helps to strengthen adaptative functions of the skin against external factors • Contributes to stimulate the differentiation and cohesion of epidermal keratinocytes • Strengthens the epidermis and increases the epidermal thickness • Supports skin integrity • Contributes to skin barrier function and reduces water loss
Applications	• Fragile Skin • Skin barrier strengthening and protection • Face and skin care • Facial/Face cleansing
Formulation data	
Concentration of use	0.1%
Solubility	Soluble in water, insoluble in oils and fats
Incorporation method	Solubilize in water up to 85°C or at room temperature for cold processing
Optimal pH	4-7



Floragenist range
Next-generation botanical extracts reinvented

The Floragenist Blossom embodies the iconic purity and mythical allure of this timeless flower, while elevating it with the precision of modern science.

Today's consumers expect authenticity from nature and clinically supported efficacy. Powered by AquaGenesis technology, our next-generation coconut-water extraction transforms each blossom bud into a concentrated elixir naturally rich in Flavanone glycosides—designed to help support skin barrier integrity and resilience.

Water derived ingredients play a central role in cosmetic science and skincare applications, contributing to skin hydration, comfort, and sensory attributes in beauty formulations. In the cosmetic and personal care industry, increasing attention has been given to naturally sourced ingredients that support skin care routines while meeting consumer expectations for sustainability and efficacy [Mendake *et al*, 2025].

Among coconut derived ingredients, coconut water has gained particular interest in recent years. Coconut water, the liquid endosperm of Cocos nucifera, is characterized by its hydrating properties and its natural content in sugars, vitamins, and minerals, which underpin its use in cosmetic and skin care applications [Mu *et al*, 2024]. Consequently, coconut water is increasingly incorporated into cosmetic formulations and beauty products, where it is valued as a refreshing, naturally derived ingredient contributing to skin and hair moisturization [Mendake *et al*, 2025].

What if we could unlock its hidden potential and harness the power of nature to create something truly extraordinary?

In 2024, 22% of US Beauty and Personal Care users reported finding it hard to trust new ingredients in products, making it essential to reintroduce familiar elements with a scientific twist [Mintel, 2024a; Mintel, 2024b]. Additionally, 36% of US facial skincare users seek natural or organic ingredients and 51% UK product launches carry natural/organic claims [Mintel, 2025a; Mintel GNPD, 2025].

This dual expectation has paved the way for next generation floral extracts powered by AquaGenesis technology designed to unlock a higher concentration of active molecules in natural extracts, combining tradition and innovation.

Let's introduce Floragenist range: aquatic elixirs obtained by AquaGenesis-patented technology using coconut water as natural solvent to enhance phytochemical content. Floragenist Blossom is part of this innovative range, alongside Floragenist Rose.

Floragenist Blossom
Orange Blossom perfected by science

Iconic and mythical Orange Blossom (*Citrus aurantium*)

Orange blossoms have carried a powerful symbolic aura, evoking purity, innocence, renewal, commitment, eternal union and divine protection.

The symbolism of the orange blossom (*Citrus aurantium*) has spanned many civilizations, from China to Europe, passing by Florida and Morocco. Bitter orange flower cultivation dates to ancient times, when the first Mediterranean civilizations discovered its aromatic and medicinal properties. The first orange tree plantations to supply perfumers were developed in south-eastern France, around Grasse, until the middle of the 20th century. But now, the bigarade orange is grown mainly in North Africa, in Tunisia, Morocco and Egypt. Orange blossom became popular at the court of King Louis XIV, accompanied by the fashion for light floral waters in the early 18th century, where "fragrant baths for ladies made from roses, lemon peel, lemon blossom, orange blossom, jasmine and laurel" are offered [Perez, 2013]. Since 1909, the orange blossom has been designated as the official flower of Florida.



A myriad of exquisite extracts to pamper and beautify = Orange Blossom Absolute.

Today, modern research reaffirms the potential of the orange blossom for skin strength. Its extracts are rich in phytochemicals, help to protect the skin from environmental damage, improved adaptive skin strength, and seals barrier integrity [Seyyedi-Mansour *et al*, 2025]. The diversity of blossom-derived products, such as Neroli Floral Water, Neroli Essential Oil, Orange Blossom Absolute, allows the formulation of skincare tailored to every skin type.

By uniting next-generation ecological innovation with the iconic purity and mythical elegance of the orange blossom, BASF scientists create cosmetic products that honor ancestral traditions while answering modern demands for sustainability and proven efficacy. The orange blossom, with its enchanting fragrance and beneficial properties, continues to be a reference in the beauty industry, offering natural and luxurious solutions to pamper the skin and reveal its inherent beauty.

Skin Strengthening: an invitation to remain strong, balanced and resilient

Strong skin is defined by a robust cutaneous barrier that maintains optimal structural and microbiological integrity, enabling it to fulfill its essential functions: limiting trans epidermal water loss (TEWL), preventing the penetration of irritants, maintaining tight intercellular cohesion, and sustaining a balanced microbiome. This robustness relies on the tightly organized architecture of corneocytes, and adequate expression of key skin integrity proteins such as Corneodesmosin, Filaggrin, Involucrin, and Desmoglein 1, all of which contribute to the skin's adaptive function by reinforcing epidermal strength and integrity [Serre *et al*, 1991]. As a result, strong skin can withstand daily environmental stressors without undergoing significant functional impairment [Wickett *et al*, 2006].

Maintaining a strong skin barrier is crucial because its resilience is not permanent. Repeated exposure to environmental aggressors—chemical irritants, climatic variations, pollution, mechanical friction, and microbial challenges—progressively weakens the barrier, shifting it toward a fragile state characterized by increased permeability and heightened reactivity making the skin less resistant to damage [Quan *et al*, 2023].

Alterations and defects of the skin barrier system at any level are often related to the development of variable skin disorders [Yosipovitch *et al*, 2019]. Disruption of epidermal homeostasis is a hallmark of skin diseases affecting over 20% of individuals worldwide . Impaired epidermal barrier function greatly contributes to both the onset and perpetuation of cutaneous inflammation. Patients with skin barrier disruption, such as atopic dermatitis, have also been shown to have higher penetration rates of a hydrophilic dye [Furuse *et al*, 2002], chemical agents such as sodium lauryl sulfate [Jakasa *et al*, 2006] or microorganisms [Son *et al*, 2014].

The barrier function can be disrupted in different ways that very often results in an increase in its permeability.

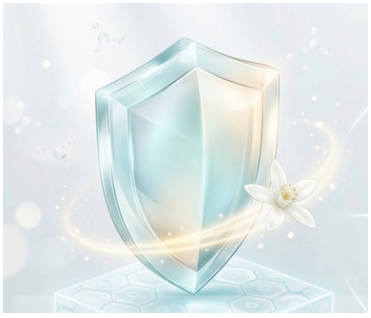
The use of cleansing products containing soaps, surfactants such as sodium lauryl sulfate (SLS) or formulas with an ethanolic base can impact the function of the skin barrier through their lipid-dissolving actions [Wickett *et al*, 2006]. The deterioration of the lipid barrier increases the risk of penetration of external pollutants, allergens and microorganisms including bacteria, and skin dehydration. Moreover, SLS-induced skin barrier defects induce altered mRNA level of keratinocyte differentiation markers such as Filaggrin (*FLG*) and Involucrin (*IVL*) and enzymes degrading corneodesmosomes [Torma *et al*, 2008].

Similarly, a compromised barrier facilitates alterations in the cutaneous microbiota and promotes the proliferation of opportunistic microorganisms such as *Staphylococcus aureus*. This bacterium interacts with Filaggrin and keratinocyte differentiation pathways, thereby exacerbating barrier dysfunction [Dahal *et al*, 2024].

Collectively, these elements highlight a central concept: even skin categorized as “strong” requires proactive reinforcement and protection, to prevent the onset of the well-established cascade involving increased permeability, inflammation, and dysbiosis.

Hidden behind skin strength: skin “integrity” proteins

Filaggrin, Desmoglein 1 and Corneodesmosin play an important role in creating a fortified, compact and cohesive *stratum corneum*.



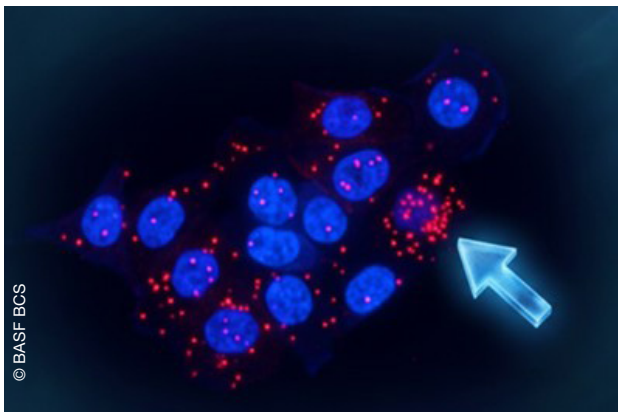
Compaction – Filaggrin (*FLG*)
Filaggrin protein facilitates the aggregation and alignment of keratin filaments within corneocytes to ensure that these cells form a robust, cohesive layer at the surface of the skin. Mutations or deficiencies in Filaggrin have been linked to various skin disorders, including atopic dermatitis and ichthyosis vulgaris, highlighting its vital importance for healthy skin [Moosbrugger-Martinz *et al*, 2022].

Cohesion – Corneodesmosin (*CDSN*)
Corneodesmosin plays a fundamental role in maintaining the cohesion of the *stratum corneum* by ensuring strong and stable adhesion between corneocytes, the outermost cells of the epidermis. By keeping these cells firmly attached, Corneodesmosin helps to create a continuous and unified skin barrier with a structural integrity that allows the skin surface to contribute to the skin's ultimate barrier and protection from external factors [Jonca *et al*, 2010].

Fortified – Desmoglein 1 (*DSG1*)
Desmoglein 1 protein is a key structural component of desmosomes (intercellular junctions) ensuring strong cell–cell adhesion in the upper epidermis. This protein plays a central role in establishing and maintaining the mechanical resilience, cohesion, and structural integrity of the skin barrier [Samuelov *et al*, 2013].

Together, Filaggrin, Desmoglein 1 and Corneodesmosin proteins have key functions on the compaction and cohesion of the *stratum corneum*, impacting skin barrier strength that helps to optimally prepare the skin for daily environmental stressors. This is why we have named them as “Skin Integrity Trio” or “skin integrity proteins”.

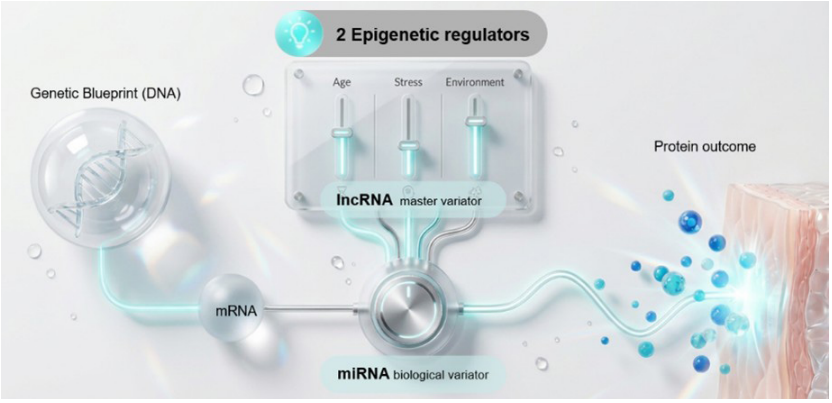
Fine-tuning skin integrity by lncRNA H19



The unseen regulator of skin integrity
It is now well established that epigenetic mechanisms play a critical role in skin health and disease. Epigenetic modifications, including DNA methylation, histone modifications, and non-coding RNA regulation, are involved in skin disorders such as psoriasis [Olejnik-Wojciechowska *et al*, 2024]. These mechanisms contribute to keratinocyte proliferation and differentiation by regulating the expression of key genes within the epidermal differentiation complex, which is essential for the terminal differentiation program of the epidermis [Moltrasio *et al*, 2022; Cohen *et al*, 2012].

So, epigenetics is an essential contributor of skin integrity by regulating the expression of genes responsible for barrier function, adapting the response to environmental stress.

Non-coding RNAs as players in skin barrier recovery
Long non-coding RNAs (lncRNAs), are unsuspected major output of genomes of complex organisms. lncRNAs are defined as non-coding transcripts of more than 200 nucleotides (200 nt), based on RNA purification protocols which allow separation of small (<200nt) and long (>200nt) molecules [Zampetaki *et al*, 2018]. In general, lncRNAs are also called ‘mRNA-like’ due to their spliced and polyadenylated composition. But we know now that it is not always the case. And lncRNAs also include circular RNAs. So, nomenclature and classification of lncRNAs is still in demand [Chodurska *et al*, 2025].



lncRNAs are often highly cell specific to a particular subcellular location, often nuclear, even if a large fraction is cytoplasmic. lncRNAs are also dynamically expressed during differentiation of mammalian cells where their expressions can be strongly influenced by environmental factors.

Like miRNAs, most lncRNAs have not been identified in genetic screens. Now, many studies have demonstrated the biological roles of lncRNAs thanks to dual genome-wide coding and lncRNA CRISPRi screens [Wu *et al*, 2022]. Such studies allowed the demonstration of their role in mammalian cell growth and migration, immune function, or epidermal homeostasis [Mattick *et al*, 2023].

LncRNA H19, a post-transcriptional regulator of keratinocyte terminal differentiation [Tang *et al*, 2020], may contribute to epidermal barrier homeostasis through its relationship with *DSG1*, a key mediator of differentiation.

DSG1, a desmosomal cadherin expressed in stratified epithelia, plays a key role in maintaining skin barrier integrity and is sensitive to alterations induced by environmental and inflammatory stressors. [Godsel *et al*, 2022; Morgner *et al*, 2023]. Factors, such as UV radiation, bacterial proteases, and irritants, can significantly reduce *DSG1* levels, thereby weakening the *stratum corneum* and increasing trans-epidermal water loss, making the skin more susceptible to allergens and pathogens [Cau *et al*, 2021].

Fortunately, advances in epigenetics offer promising strategies to counteract this decline. One key player is the lncRNA H19, which has been shown to promote *DSG1* expression, especially during cellular differentiation. Silencing lncRNA H19 leads to reduced *DSG1*, whereas ectopic (i.e., experimentally induced) expression of lncRNA H19 can boost *DSG1* levels [Li *et al*, 2017]. Harnessing epigenetic mechanisms, particularly those involving lncRNA H19, could therefore help strengthen the skin barrier by enhancing *DSG1* production.

Cellular localization and expression level of the epigenetic lncRNA H19 in epidermal differentiation

Involucrin expression, which is initiated early during keratinocyte differentiation and persists throughout terminal differentiation, is widely used as a marker to assess the induction of keratinocyte differentiation, including in calcium-induced models [Crish *et al*, 2002; Hahn *et al*, 2016].

This study investigates the co-expression of *lncRNA* H19 and its target *DSG1* under differentiation process triggered by calcium stimulation and monitored by *IVL* induction.

The co-increase of lncRNA H19 (red) and *IVL mRNA* (yellow) can be visualized in Figure 1 after keratinocyte differentiation induced by calcium treatment. Moreover, the boost of the lncRNA H19 target *DSG1* was quantified by RT-qPCR (Fig. 2).

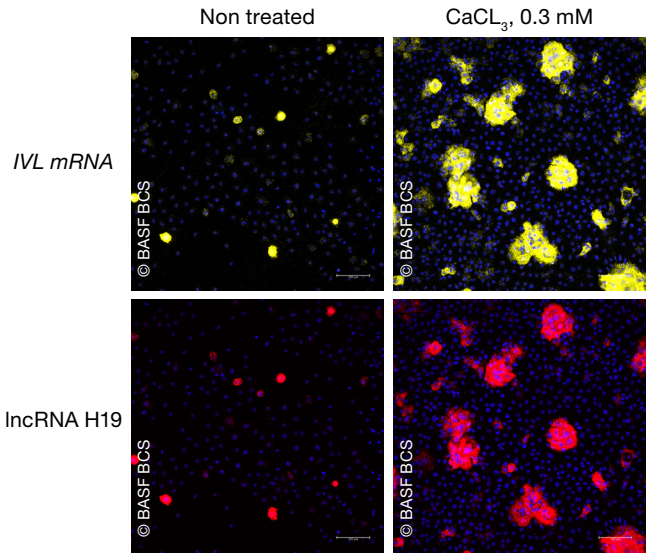


Figure 1 - Illustrative pictures of staining of *IVL* mRNA (yellow) and lncRNA H19 (red) with or without calcium treatment to increase differentiation.

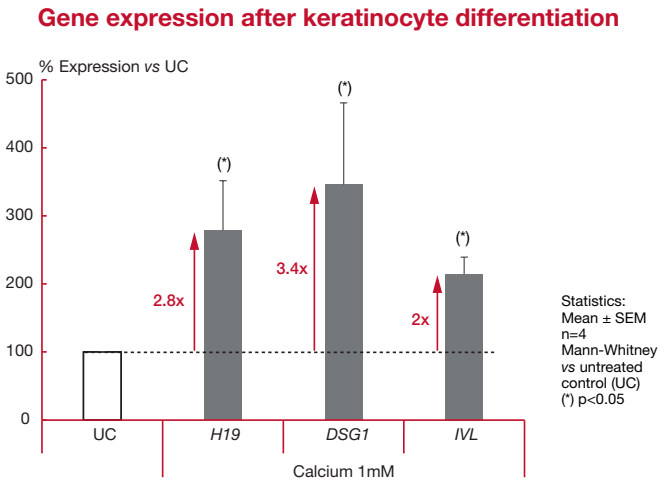


Figure 2 - Gene expression of lnc-H19, *DSG1* and *IVL* after keratinocyte differentiation stimulation in keratinocyte culture.

MATERIALS&METHODS

Study numbers

TME2021-008-2024-BM-P02-E03
EMB180601-2022-770-BM 2025

lncRNA visualization

After 48h of human keratinocyte culture with or without calcium 0.3 mM and fixation, slides were washed twice in PBS and then dehydrated by successive 5 minutes incubations in ethanol at 50%; 70%; and twice at 100% before drying at room temperature.

In situ hybridization were performed using the RNAscope Plus smRNA-RNA HD Reagent Kit (ACD Biotechne), according to the manual instructions. Shortly, endogenous peroxidase was inactivated by incubation with Hydrogen Peroxide solution for 10 minutes at room temperature and then cells were permeabilized by Protease III treatment for 30 minutes at room temperature before 2 PBS washings. Then, specific probes against lncRNA H19 and *IVL* mRNA were incubated on cells for 2 hours at 40°C before 2 buffer washings.

Signal amplification was performed for each probe using a 3 steps revelation, the last one corresponding to a specific fluorochrome (TSA Vivid Florochrome 650 and TSA Vivid Florochrome 570 for H19 or *IVL* revelation, respectively). After washing, cell nuclei were labelled by DAPI and then slides mounted. Fluorescent image captures were performed using a Leica SP8 microscope.

RNA extraction and analysis

Long and short RNAs from human keratinocyte cultures with or without calcium treatment at 1 mM for 48h, were extracted with NucleoSpin miRNA kit (Machery-Nagel), according to the manual instructions. Quantification and quality control of RNAs were performed by measurement of optical densities at 260 and 280 nm.

qRT-PCR for lncH19 and gene level measurement

Reverse transcriptions were realized on 0,1 µg of long RNA in 20 µl final mix using SuperScript IV VILO Master Mix Kit (Invitrogen) and the C100 Touch Thermocycleur (BioRad) for each condition.

Quantitative PCRs were performed on 2 µl of RT mix, diluted at 1/2, according to manual instruction (LC 480 SYBR Green I Master, Roche) in a LC480 thermocycler (Roche).

Each strain analysis was performed in duplicate and the mean used. Results were normalized compared to the mean of 2 housekeeping genes analyzed in duplicate: ACTB and SDHA (coding for Actin B and succinate dehydrogenase subunit A, respectively). For each target, the gene expression modulation was calculated using the ΔΔCt method.

Results and statistics

The results are expressed as mean and percentage ± SEM (n=4) compared to untreated control condition (UC) standardized to 100%. The statistical analysis was done using Mann-Whitney Rank Sum test *vs* UC, with a threshold of significance set at 5% ($p<0.05$).

Traditional and organic *Citrus aurantium*

Natural extract of *Citrus aurantium* flowers sourced from sustainable and ethical floral culture.

Description of the plant

Family: Rutaceae

Botanical name: *Citrus aurantium var amara*

Vernacular name: *Citrus aurantium amara* (flowers); *Citrus aurantiifolia* (Christm.) Swingle; *Citrus aurantium* var. amara L.; *Citrus aurantium* var. bigaradia Hook. f.; *Citrus aurantium* var. voangkely H. Perrier; *Citrus bigaradia Risso* & Poit.; *Citrus hystrix* H. Perrier; *Citrus vulgaris Risso.*

Synonym: *Citrus bigaradia*

Description & Distribution of the plant: bitter orange, Seville orange, sour orange, bigarade orange, or marmalade orange is the citrus tree *Citrus aurantium* and its fruit. It is native to Southeast Asia and has been spread by humans to many parts of the world.

Habitat: the tree thrives in sunny climates with moderate humidity, and grows best in well-drained loamy soils, enriched with compost and sand. Waterlogged soil should be avoided, as they are harmful to root health.

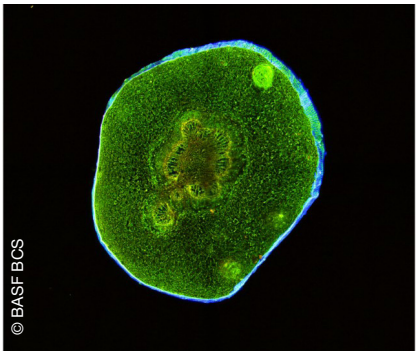


Figure 3 - Histological localization of polyphenols in cross section of orange blossom peduncle, using Neu's reagent. Spectral study using confocal microscope. Flavanone glycosides (green).

Spectral analysis of phenolic compounds using confocal microscopy in sections of *Citrus Aurantium* flower peduncles.

The image (Fig. 3) reveals the localization of phenolic compounds in the orange blossom peduncle.

MATERIALS&METHODS

Flower peduncle cross-sections (100µm) were obtained using a blade hand microtome (Reference : 574082 Jeulin, France) and then dipped in Neu then Polyethylene glycol and glycerol/water (50:50) solution.

A spectral study was conducted using a confocal microscope (Leica Microsystems SAS, Nanterre, France) to characterize the fluorescence absorption and emission properties of Flavanone glycosides (green).

Origin: Morocco

Part used: flower buds



Sourcing and harvest: the *citrus aurantium* is cultivated in Morocco, in the regions of Marrakech and Sidi Allal Tazi, in several farms that are certified organic and committed to social and environmental initiatives.

The sourcing approach of Floragenist Blossom supports the goals of BASF's.

Responsibly active program for its cosmetic bioactives portfolio.

For more information about the sourcing, please refer to Botanical sourcing information available upon request.

Traditional and modern uses

Bitter orange blossom occupies a special place in Moroccan traditions and rituals. Used for its soothing and purifying virtues, it is often incorporated into religious and festive ceremonies. Orange blossom water, obtained by distillation, is used to bless weddings, births, and other important family events [Tavares *et al*, 2022; Gharaghani *et al*, 2026].

AquaGenesis technology: use of coconut water as natural solvent for enriched plant extracts

The use of *Cocos nucifera* fruit for health and beauty

The fruit of the coconut palm (*Cocos nucifera* L.) has probably been used by humans for centuries. Nearly one-third of the world's population depends, to some extent, on coconuts for their food and economy. In these cultures, coconuts have a long and respected history [Obidoa *et al*, 2009]. Coconut is classified as a "functional food" because it offers many health benefits, beyond its nutritional content. The edible endosperm is used either as a solid white kernel or as a clear nutrient liquid (coconut water) [Prades *et al*, 2012a; Prades *et al*, 2012b]. In particular, the coconut water is known to possess a range of health-promoting nutrients, and it is commonly used for food, pharmaceutical or cosmetic applications [Kumar *et al,* 2021].

New use of *Cocos nucifera* fruit for bio-based technologies

Aqueous extraction, commonly used in the manufacture of plant-based cosmetics, is considered as safe and environmentally friendly method. Research by BASF and the University of Avignon and the Vaucluse region (in France) has shown that coconut water can be used as a natural solvent to improve plant extraction, thereby enhancing the phytochemical content [Makerri *et al*, 2018]. This patented technology, offers new benefits for the cosmetics industry.

The coconut water is sourced from organically cultivated dwarf coconut trees in Tamil Nadu, India, and is hand-harvested year-round. Extracted from young green coconuts aged 6–9 months, the entire fruit is valorized to minimize waste.

What brings this patented extraction process?

AquaGenesis is an extraction technology that uses natural coconut water as solvent to extract rose flowers. Compared with aqueous extraction, it enriches the phytochemical compounds of the extract.

Controlled particle size and porosity of botanical raw material (= matrix) influence extraction efficiency while preserving process robustness. Temperature control accelerates diffusion and solubility without compromising molecular integrity [Chemat F, Rombaut N, *et al*, 2015].

An extraction solvent is characterized by different physico-chemical parameters and the key relevant ones for the use of coconut water as extraction solvent are the following: pH, osmolarity, ionic strength and surface tension [Yong *et al*, 2009].

Since surface tension belongs to the physical parameters that is directly linked to the enhanced penetration of solvents into plant tissues, facilitating the release of phytochemical compounds [Samant *et al*, 2025], we compared coconut water and water as solvents based on this property.

Indeed, the surface tension of water at 20°C is well described in literature (≈ 72 $mN{\cdot}m^{-1}$), data that was confirmed by internal measures. The presence of sugars and electrolytes reduces this value of 33 %) for coconut water at the same temperature. With a decreased surface tension of the solvent referring to water, the wetting of the matrix (grinded rose flowers) is enhanced allowing better solvent access. This results in a more efficient diffusion of targeted phytochemical compounds of rose flower with the coconut water

Benefits of AquaGenesis extraction process on Floragenist Blossom extraction

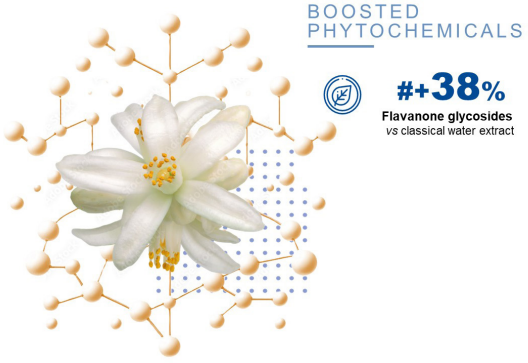


Figure 4 - Enriched phytochemicals of blossom extract with AquaGenesis technology process *vs* water.

We have performed a phytochemical comparison of similar orange blossom extracts obtained using Coconut water or water. The results revealed a similar molecular profile with an enrichment in Flavanone glycosides (+38% *vs* aqueous extraction) in the extract obtained with Coconut water as solvent (Fig. 4).

Additionally, *in vitro* investigations on *CDSN* protein synthesis in human keratinocytes demonstrated that Floragenist Blossom tends to outperform water-*Citrus aurantium* extract, increasing corneodesmosin expression by +30% (*vs* aqueous extraction process) the level of the surface protein Corneodesmosin known for strengthening cell cohesion within the *stratum corneum*.

Powered by the innovative AquaGenesis extraction technology, this *Citrus aurantium* flower extract is optimized to enhance its phytochemical content and support increased expression of the cohesion protein Corneodesmosin.

MATERIALS&METHODS

Tested products

Coconut-water-*Citrus aurantium* extract (INCI: Cocos Nucifera (Coconut) Fruit Extract (and) Citrus Aurantium Amara (Bitter Orange) Flower Extract) is the "excipient free" part of Floragenist blossom.

Coconut-water–*Citrus aurantium* Extract (batch CS220321.01; INCI: Cocos Nucifera (Coconut) Fruit Extract (and) Citrus Aurantium Amara (Bitter Orange) Flower Extract)) and freeze-dried water–*Citrus aurantium* extract (batch EXT220921.01) were prepared with similar content of Citrus raw material.

Control coconut water solvent without Citrus raw material (batch EXT220524.01) was used to confirm that the quantified Flavanone glycosides originated from the flowers (data not shown). Moreover, the control coconut water was used for surface-tension investigations.

Flavanone glycoside measurement

Flavanone glycosides were quantified by high-performance liquid chromatography coupled to a photodiode array detector (HPLC–PDA) using an Ultimate 3000 system (Thermo Scientific Dionex) equipped with a PDA detector. The following compounds were identified: Neoeriocitrin, Narirutin, Naringin, Hesperidin, Neohesperidin.

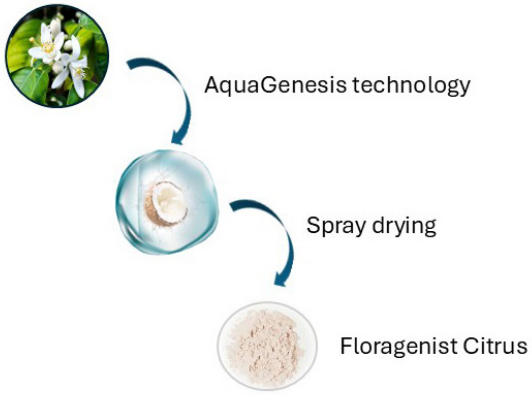
Quantitative analysis was performed using an external calibration approach. Reference standard of Naringin was used to generate calibration curves (purity assumed to be 100%).

The total Flavanone glycoside content was calculated as the sum of Neoeriocitrin, Narirutin, Naringin, Hesperidin and Neohesperidin presented in Figure 6B and expressed in Naringin equivalents.

***CDSN* protein**

The cellular Corneodesmosin was measured by a specific commercial ELISA method. Briefly, 100 µL of standard or sample was added to each well of the microplate and incubated for 2.5h at room temperature. After washing, 100 µL of the biotinylated antibody was added to each well and incubated for 1h at room temperature. After washing, 100 µL of Streptavidin solution was added and incubated for 45min at room temperature with gentle shaking. After washing, 100 µL of TMB One-Step Substrate Reagent was added to each well and incubated for 30min at room temperature in the dark with gentle shaking. Finally, 50 µL of Stop Solution was added to each well, and absorbance was measured immediately at 450 nm using a microplate reader.

Processing



Floragenist Blossom is obtained through patented AquaGenesis extraction process that uses coconut water as the solvent, applied to dried and crushed Citrus aurantium flower buds (Fig. 5). After removing insoluble matter, the extract is concentrated and transformed into a spray-dried, preservative-free powder with maltodextrin. Floragenist Blossom is characterized by its content in Flavanone glycoside (≥ 4.0%).

Figure 5 - Processing.

Composition

The typical composition of coconut-water-*citrus-aurantium* extract is detailled in table 1.

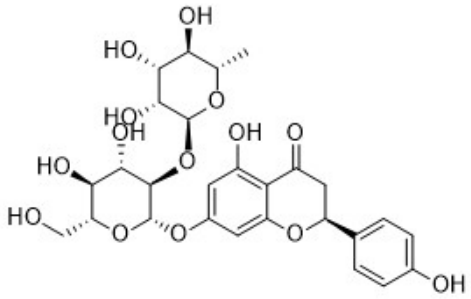
Among the principal constituents extracted from the *Citrus Aurantium* flower, Flavanone glycosides are identified by LC PDA using Naringin as the reference compound (Fig. 6B)

Carbohydrates are quantified by total sugars analysis using the colorimetric phenol method. In addition, protides (amino acids, peptides & proteins) are determined by measuring total nitrogen via the Kjeldhal method, applying a conversion factor of 6.25. Furthermore, minerals are measured through ash measurement by calcination.

Components	Range %
Carbohydrates	42.8 - 72.8
Minerals and salts	3.98 - 7.78
Protides (amino-acid and proteins)	5.37 - 10.38
Flavonols glycosides	11.60 - 14.18

Table 1 - Typical phytochemical composition of Coconut-water-*Citrus-aurantium* extract (*Cocos Nucifera* (Coconut) Fruit Extract (and) *Citrus Aurantium* Amara (Bitter Orange) Flower Extract) in Floragenist (weight % on extract, based on 6 representative batches).

A



Floragenist Blossom is characterized by its sum of identified flavanones glycosides (Fig. 6B) expressed in Naringin equivalent (>= 4%). Among them, naringin (Fig. 6A) was identified by liquid chromatography (LC MS).

B

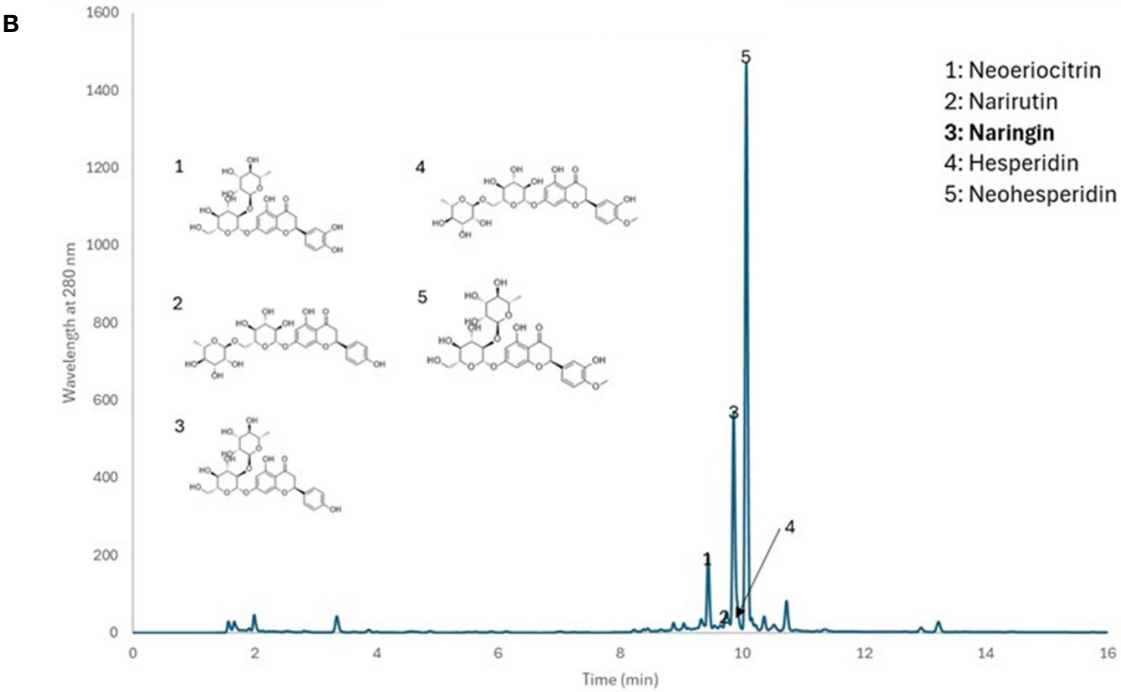


Figure 6 - **A**- Naringin structure and **B**- example LC PDA typical chromatogram for Floragenist Blossom.

Naringin (Fig. 6A), a Flavanone-7-O-glycoside, is a water-soluble secondary metabolite belonging to the flavonoid family.

Naringin is the predominant flavonoid found in Citrus fruits, which are widely consumed during winter in the Mediterranean diet [Tripoli, 2007].

Sensory evaluations indicate that naringin, at concentrations typically found in grapefruit juice or citrus peels, imparts a strong and persistent bitterness [Danton *et al*, 2018]. According to the literature, naringin has been widely studied for its pharmacological and therapeutic properties, notably cardioprotective, hepatoprotective, anti-diabetic and anti-hyperlipidemic effects [Viswanatha *et al*, 2017].

Experimental studies have also demonstrated that naringin exhibits anti-inflammatory, antioxidant, and anti-apoptotic activities [Yao H *et al*, 2018].

Safety / tolerability of the product

Floragenist Blossom was evaluated to ensure its safety in the recommended conditions of use.

Description

Floragenist Blossom is a coconut water extract obtained from organic Citrus aurantium flowers, obtained through the patented AquaGenesis technology, titrated in Flavanone glycoside, available as a preservative-free powder.

INCI name: Maltodextrin (and) Cocos Nucifera (Coconut) Fruit Extract (and) Citrus Aurantium Amara (Bitter Orange) Flower Extract

Naturalness content (according to ISO 16128): 100% from natural origin

Recommended dose of use: 0.1%

China IECIC listed

Approved by ECOCERT RAW MATERIAL COSMOS APPROVED



Example: illustration of commercial samples of Floragenist Blossom with a hydrogel and emulsion containing 0.1% of Floragenist Blossom.



Demonstrated efficacy

Maintaining a strong skin barrier requires a tightly organized corneocyte architecture and adequate expression of key skin integrity proteins—such as Corneodesmosin (*CDSN*), Filaggrin (*FLG*), Involucrin (*IVL*), and Desmoglein 1 (*DSG1*)—to sustain intercellular cohesion and help limit skin permeability.

Floragenist Blossom was assessed in reconstructed epidermis, keratinocyte-based models, *ex vivo* skin explants, and a placebo-controlled split-face study, using quantitative readouts of barrier function (e.g., TEWL and permeability), epidermal morphology, and molecular markers of differentiation/cohesion, including lncRNA H19 and selected barrier-related proteins/genes.

In vitro, the properties of Floragenist Blossom were evaluated on:

- The increase in epidermal thickness in reconstructed epidermis.
- The reduction of skin permeability induced by SLS-stress in reconstructed epidermis.
- The enhanced restoration of Filaggrin (FLG) expression following mechanical stress in *ex vivo* model .
- The enhance of the adaptive protection of FLG under bacterial disturbance in 2D model.
- The protection of skin barrier (CDSN in protein on keratinocytes).
- The protection of skin barrier integrity (lncRNA H19, *DSG1* and *IVL* in keratinocytes).
- The stimulation of 12 genes involved in cohesion and differentiation in keratinocytes.

In vivo, TEWL was recorded at baseline (pre-challenge) and after a standardized SLS challenge following 28 days of Floragenist Blossom application, to quantify changes in barrier disruption response *vs* placebo.

EFFICACY

Epidermal thickness increase

In vitro

OBJECTIVE

Epidermal aging is characterized by a progressive thinning and flattening of the epidermis, which is described as a major structural change in aged human skin. This epidermal thinning is explicitly associated with impaired barrier function and increased susceptibility to mechanical shear stress. As a result, aged skin shows greater vulnerability to environmental aggressors and reduced protective capacity [Quan *et al*, 2023].

So, the aim of the study was to investigate the effect of Floragenist blossom on the morphogenesis of the epidermis. The thickness of the epidermis was measured on a 3-dimensional model of reconstructed epidermis.

RESULTS & DISCUSSION

The effect of Floragenist Blossom on the thickness improvement of the epidermis was illustrated in Figure 7 with 2 tested-concentrations on reconstructed epidermis after 5 days of treatments. We observed a significant increase in epidermal thickness by 32% and 59% (± SEM 9 & 10, $p<0.001$, *vs* Untreated control normalized at 100% SEM 9) after 5 days of treatments by respectively 0.02 and 0.06%.

CONCLUSION

Floragenist Blossom significantly increases the thickness of the epidermis in the 3D model of reconstructed human epidermis and could help to improve the quality of the skin barrier.

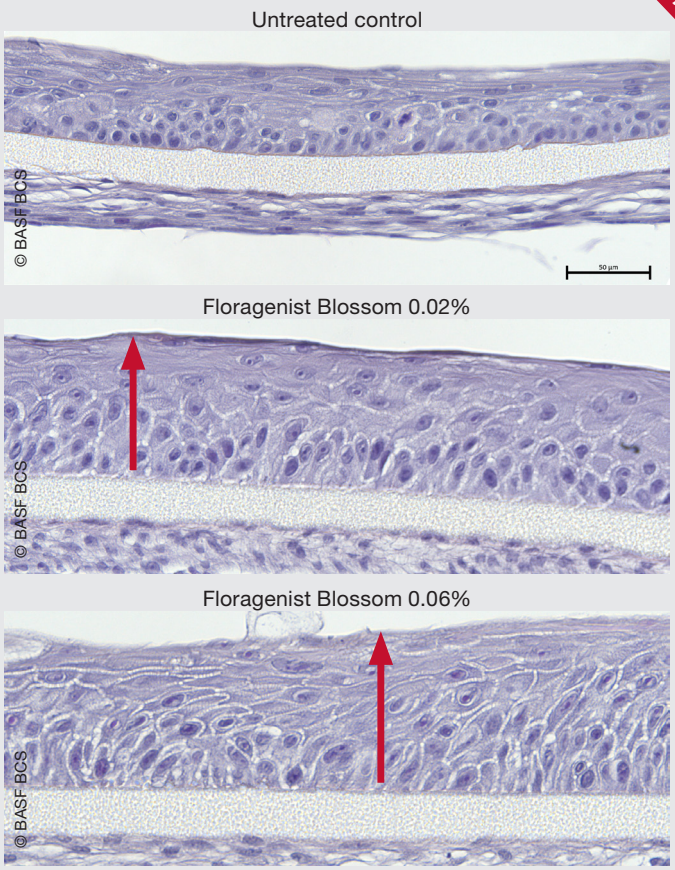


Figure 7 - Illustrative pictures skin barrier strengthening in reconstructed epidermis visualization of the epidermal thickness after treatment with Floragenist Blossom at 0.02% and 0.06% for 5 days.

MATERIALS&METHODS

Study number

EMB180601-2022-692-MEXP_RHE1

Tested products

Coconut-water-Citrus-aurantium extract (batch JVEGID3194AA 6GAPEID06900; INCI: Cocos Nucifera (Coconut) Fruit Extract (and) Citrus Aurantium Amara (Bitter Orange) Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Blossom at 0.02 and 0.06%.

Reconstructed human epidermis Treatments

Reconstructed human epidermis (RHE) models were cultured for an initial period of 4 days to promote keratinocyte proliferation. Following this, the cultures were transferred to differentiation medium and maintained at the air-liquid interface for 12 days to allow maturation of the epidermal tissue. During this maturation phase, the culture media was refreshed every two days to ensure optimal conditions for tissue development. Multiple experimental conditions were assessed during this period. An untreated control group, consisting of reconstructed epidermis cultured exclusively in differentiation medium and without any additional treatment, was terminated after 10 days of maturation for subsequent analysis. After 6 days of differentiation, experimental treatments with or without Floragenist Blossom at 0.02 and 0.06% were initiated for a period of 4 days.

Epidermis thickness

For each experimental condition, a total of nine hematoxylin and eosin (H&E)–stained histological images were acquired. Images were obtained from three independent reconstructed epidermis blocks per condition, with three non-overlapping representative fields captured from each block. All images were acquired using identical magnification and acquisition settings to ensure consistency across samples.

Epidermal thickness quantification was performed by measuring the distance from the basal layer to the cornified layer. For each image, six independent thickness measurements were taken at evenly distributed positions across the epidermis to account for local variability. This resulted in a total of 54 individual measurements per experimental condition.

Image analysis was conducted using FIJI software (ImageJ2, open-source). Prior to measurement, images were calibrated using the corresponding scale bar. Quantitative data were subsequently exported and analyzed using SigmaPlot version 12.5. Mean values and standard deviations were calculated for each condition, and statistical comparisons between experimental groups were performed using standard analytical protocols.

Results and statistics

The results are expressed as mean and percentage ± SEM (n=7 to 9) compared to untreated control condition (UC) standardized to 100%. The statistical analysis was done using OneWay ANOVA and Dunnett's *vs* UC, with a threshold of significance set at 5% ($p<0.05$).

EFFICACY

Reduction of skin permeability induced by SLS stress

in vitro

OBJECTIVE

The skin barrier minimizes trans-epidermal water loss (TEWL) and preserves internal hydration by restricting diffusion and keeping TEWL within healthy physiological ranges. When the epidermal barrier is compromised by a chemical agent, such as sodium lauryl sulfate (SLS), permeability is increased [Jakasa *et al*, 2006].

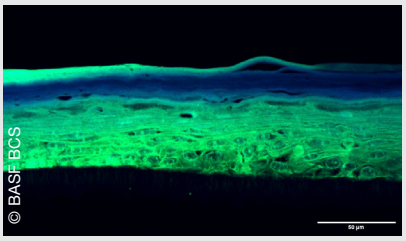
So, the aim of this study was to evaluate the skin barrier protection potential of Floragenist Blossom by measuring the penetration of a fluorescent probe after SLS stress on a reconstructed human epidermis.

RESULTS & DISCUSSION

Models with SLS stress showed a significant increase in Lucifer Yellow (LY) fluorescent dye penetration in the reconstructed epidermis, illustrating the disrupted barrier (Fig. 8).
Fluorescent dye increased by 94% when stressed by SLS 0.01% (SEM 5%, Mann-Whitney Rank Sum Test, $p<0.001$ *vs* the untreated control at 6%, SEM 3%).

Moreover, Floragenist Blossom 0.02% in presence of SLS 0.01% demonstrated a significant decrease of the LY fluorescent dye penetration in the reconstructed epidermis by 40% after 4 days of treatments (SEM 6%, Mann-Whitney Rank Sum Test, $p<0.001$ *vs* the SLS stress normalized at 100%, SEM 5%).

Fluorescent probe + SLS at 0.01%



Fluorescent probe + SLS at 0.01% + Floragenist Blossom at 0.02%

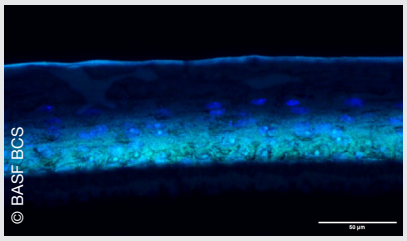


Figure 8 - Illustrative pictures of barrier protection by Floragenist Blossom in presence of SLS stress. Increase of fluorescent probe retention in epidermis after barrier disfunction induced by 0.01% SLS stress on reconstructed human epidermis.

CONCLUSION

Floragenist Blossom reduced the penetration of the fluorescent probe into the epidermis after skin barrier disruption by SLS and could help improve the integrity of the skin barrier.

MATERIALS&METHODS

Study number
EMB180601-2022-744-MEXP-RHE2

Tested products
Coconut-water-Citrus-aurantium (batch PG22083.01; INCI: Cocos Nucifera (Coconut) Fruit Extract (and) Citrus Aurantium Amara (Bitter Orange) Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Blossom at 0.02%.

Reconstructed epidermis and treatments
Reconstructed human epidermis (RHE) models were cultured for an initial period of 4 days to promote keratinocyte proliferation. Following this, the cultures were transferred to differentiation medium and maintained at the air-liquid interface for 12 days to allow maturation of the epidermal tissue. During this maturation phase, the culture media was refreshed every two days to ensure optimal conditions for tissue development. Multiple experimental conditions were assessed during this period. An untreated control group, consisting of reconstructed epidermis cultured exclusively in differentiation medium and without any additional treatment, was terminated after 10 days of maturation for subsequent analysis. After 6 days of differentiation, experimental treatments with or without Floragenist Blossom were initiated for a period of 4 days.

Subsequently, 0.01% sodium lauryl sulphate (SLS) was applied to the surface of the reconstructed epidermis for 16 hours from day 10 of maturation. This served as a positive control to induce disruption of the epidermal barrier. All experimental conditions were conducted in triplicate to ensure statistical reliability and reproducibility of results.

Lucifer yellow penetration test
RHE samples were fixed in 4% paraformaldehyde at 4°C for 24 hours, subsequently embedded in paraffin, and sectioned at a thickness of 5 µm. Following deparaffinization and rehydration, sections were mounted using Prolong Gold mounting medium containing DAPI, coverslipped, and imaged. Lucifer yellow fluorescence was detected utilizing a Zeiss LSM 700 confocal microscope (excitation wavelength: 428 nm; emission wavelength: 536 nm) and quantitative analysis was performed with FIJI software (Image J2, open source).

Results and statistics
The results are expressed as mean and percentage ± SEM (n=10 to 16) compared to the stressed condition with SLS 0.01% standardized to 100%. The statistical analysis was done using Mann-Whitney Rank Sum Test *vs* SLS stress, with a threshold of significance set at 5% ($p<0.05$).

EFFICACY

Filaggrin recovery after mechanical stress

ex vivo

OBJECTIVE

Loss of the superficial layers of the epidermis also contributes to the loss of the integrity proteins and thus to the loss of the barrier function through the decrease of Filaggrin (FLG) [Dang *et al*, 2014].

So, the aim of this study was to evaluate the effect of Floragenist Blossom on the Filaggrin recovery after mechanical stress by stripping, in skin explants, using immunostaining assay.

RESULTS & DISCUSSION

Stripping of the superficial layers of the epidermis demonstrated an immediate decrease of Filaggrin by 23% (± SEM 3% *vs* baseline (D0) 100% ± 5) (Fig. 9A) which can consequently affect the barrier function.
Floragenist Blossom 0.02% treatment (Fig. 9B) significantly improved Filaggrin recovery by 2.5x time (± SEM 11%, $p<0.001$) *vs* stripping control (D5) normalized at 100% (SEM 9%).

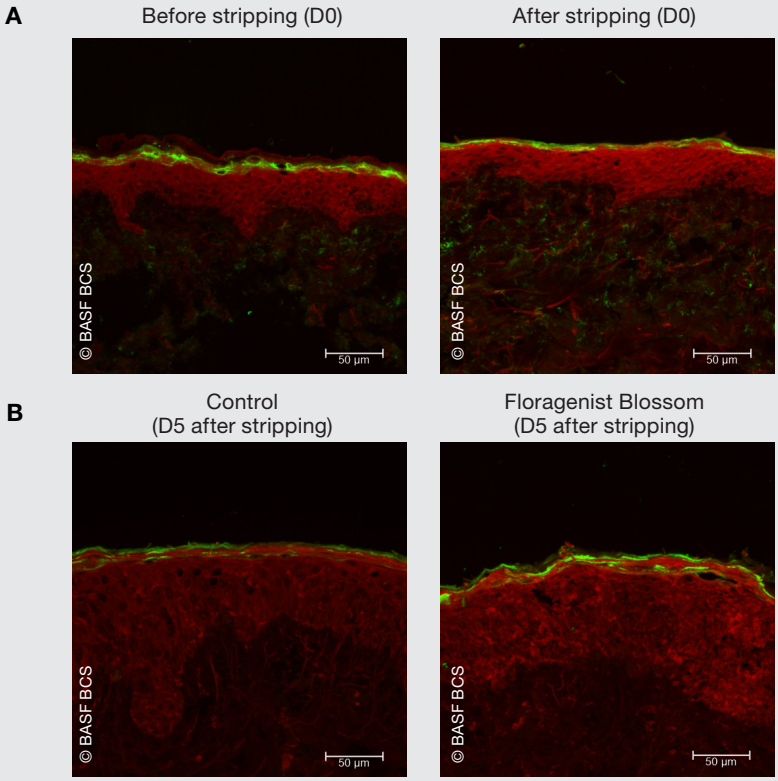


Figure 9 - **A**- Illustrative pictures of the decrease of the presence of Filaggrin in the superficial layers of the epidermis in skin explants after stripping at D0. **B**- illustrative picture of the increase of the presence of FLG with stimulating effect following 5 days of Floragenist Blossom treatment at 0.02% after the stripping step.

CONCLUSION

Floragenist Blossom significantly improved filaggrin recovery previously lost with stripping, helping to recover the barrier after mechanical stress.

MATERIALS&METHODS

Study number
EMB180601-2023-783-HI

Tested products
Coconut-water-Citrus-aurantium (PG22083.01; INCI: Cocos Nucifera (Coconut) Fruit Extract (and) Citrus Aurantium Amara (Bitter Orange) Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Blossom at 0.02%.

Skin explants & treatments
Human skin biopsies were collected from a 60-year-old female donor and processed under sterile conditions. The hypodermis was carefully removed using a sterile scalpel, and the epidermal surface was dried with sterile gauze. To disrupt the *stratum corneum*, the biopsies underwent 40 rounds of tape stripping. Afterwards, each skin explant was cut into 1 × 1 cm^2 pieces and washed five times (10 minutes each) in a solution containing 5% Gentamicin and 1% antifungal agent, followed by a final rinse in phosphate-buffered saline (PBS) to minimize contamination.
For culture, biopsies were placed in Transwell inserts (Costar, Ref. 3412; 24 mm diameter, 0.4 µm polycarbonate membrane) within 6-well plates, each well containing 2 mL of Williams E medium supplemented with antibiotics. Floragenist Blossom was topically applied at 0.02% during 5 days after stripping at D0. Each biopsy received 10 µL of the designated treatment solution on the surface. Cultures were maintained at 37°C with 5% CO_2 and over 95% relative humidity. All samples were harvested on day 5 for analysis.

FLG protein immunostaining
Following harvest, samples were embedded in OCT compound, sectioned at 10 µm thickness for histological analysis. Cryosections were fixed in 4% paraformaldehyde or acetone for 20 minutes at room temperature (RT), rinsed in PBS, and permeabilized with 0.2% Triton X-100 in PBS for 15 minutes. After further PBS washes, sections were blocked with 2% serum for 2 hours at RT.
For Filaggrin staining, sections were incubated with a primary antibody against filaggrin (Abcam, ab218395) diluted 1:300 in blocking buffer, for 2 hours at RT.
After primary antibody incubation, sections were washed in PBS and incubated with Alexa Fluor 488-conjugated secondary antibody (1:200 dilution) for 45 minutes at RT.
Evans Blue was used for counterstaining to highlight tissue morphology. Slides were then rinsed thoroughly in PBS.
Stained sections were subsequently mounted using VECTASHIELD mounting medium containing DAPI, followed by cover-slipping and imaging with a Leica TCS SP8 confocal microscope.
Filaggrin immunoreactivity was detected using the Alexa Fluor 488, excited at 488 nm with emission collected between 500–550 nm (green). Tissue morphology was highlighted by Evans Blue fluorescence, excited at 633 nm with emission collected above 650 nm (red).
For each treatment condition, six representative images were acquired under identical imaging parameters. Filaggrin expression was quantified by measuring the integrated fluorescence intensity of the Alexa Fluor 488 signal using FIJI software (ImageJ2, open-source). Regions of interest corresponding to the epidermal layer were consistently defined across samples, and background fluorescence was subtracted prior to analysis. Quantified values were averaged per condition for subsequent statistical evaluation.

Results and statistics
The results are expressed as mean and percentage ± SEM (n=4 to 6) compared to the untreated control (UC) corresponding to the condition with only stripping at D0, standardized to 100%. The statistical analysis was done using Student t test *vs* UC, with a threshold of significance set at 5% ($p<0.05$).

EFFICACY

Filaggrin protection from dysbiosis

In vitro

OBJECTIVE

In 90% of atopic dermatitis, *S aureus* is significantly induced compared to healthy skin which greatly affects the expression of protein-coding genes including Filaggrin and Loricrin [Son *et al*, 2014].

So, the aim of this study was to evaluate the efficacy of Floragenist Blossom on the expression rate of Filaggrin on keratinocytes in the presence of *Staphylococcus aureus* bacterium.

RESULTS & DISCUSSION

In keratinocyte culture model, the presence of *S. aureus* decreases significantly the expression of the terminal differentiation markers Filaggrin (Fig. 10). In presence of *S. aureus,* Filaggrin demonstrated a decrease of 61% (± SEM 9% *vs* untreated control 161% SEM 12) (Fig. 10). Moreover, Floragenist Blossom, evaluated at 0.02% in this model, showed an increase of Filaggrin protein level after 5 days by 2.7 times (± SEM 43% *vs S. aureus* stress normalized at 100%, SEM 9%, p<0.001).

A - Untreated control | **B** - *S. aureus* | **C** - *S. aureus* + Floragenist Blossom 0.02%

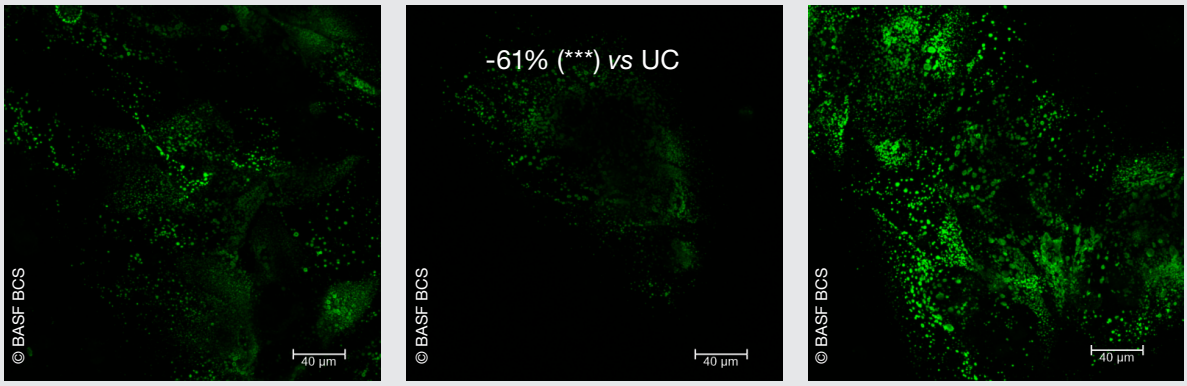


Figure 10 - Illustrative pictures of Filaggrin protein (in green) decreased in keratinocyte culture with *S. aureus* (**B**) compared to baseline (**A**) and increased by Floragenist Blossom treatment (**C**) in keratinocytes.

CONCLUSION

Floragenist Blossom significantly improved Filaggrin recovery previously lost after *S. aureus* stress helping to protect the barrier from bacterial disturbance.

MATERIALS&METHODS

Study number
EMB180601-2022-699-HI_aureus

Tested products
Coconut-water-Citrus-aurantium (batch JVEGID3194AA 6GAPEID06900; INCI: Cocos Nucifera (Coconut) Fruit Extract (and) Citrus Aurantium Amara (Bitter Orange) Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Blossom at 0.02%.

***S. aureus* stress and treatments**
Human keratinocytes were seeded on slides (LabTek II, Nunc) at 80.000 cells/cm² on a type I collagen coating and cultured in a culture medium MCDB153 added 2% fetal bovine serum at 37°C, 5% CO_2, relative humidity >95% for 5 days. *Staphylococcus aureus* ATCC 35556 bacteria are cultured in a TSB-F broth (Biomérieux) to have 1x10^7 bacteria/ml of TSB-F broth. The bacteria are killed at 90°C for 15 minutes. These are then brought into contact with keratinocytes at a rate of 2x10^6 bacteria per cm². Keratinocytes in contact with the killed bacteria were treated with the products and cultured in MCDB153 culture medium in a controlled atmosphere for an additional 6 days.

Filaggrin protein immunostaining
The cells are fixed with acetone at -20°C for 10 minutes and rinsed with PBS. Immunostaining was performed using a primary anti-Filaggrin antibody (Abcam ab218395) and a secondary antibody coupled to fluorochrome Alexa 488 (Fisher Scientific, A21121) after saturation of non-specific sites with 2% serum. Samples are observed by confocal microscopy (Leica SPE). The expression of Filaggrin was quantified by image analysis using the FIJI software (Image J2, open source).
The results were normalized to the control in the presence of *S. aureus* and expressed as the mean (%) ± standard error of the mean (SEM) over 6 images per condition and 3 independent experiments. Statistical analysis was performed using the student t-test or Mann-Whitney test compared to control in the presence of *S. aureus* or Coconut-water-*Citrus-aurantium* extract. Differences are considered significant for a p-value<0.05.

Results and statistics
The results are expressed as mean and percentage ± SEM (n=18) compared to the stressed condition with S aureus stress, standardized to 100%. The statistical analysis was done using Student t test or Mann-Whitney Rank Sum Test *vs S. aureus* stress, with a threshold of significance set at 5% (p<0.05).

EFFICACY

Stimulation of Corneodesmosin protein on keratinocytes

In vitro

OBJECTIVE

Corneodesmosin supports the formation of a seamless, cohesive skin barrier, maintaining the cohesion of the *stratum corneum* by ensuring strong and stable adhesion between corneocytes. reinforcing its structural integrity so that the skin surface can fully contribute to optimal barrier function and protection against external aggressors [Haftek 2015].

So, the aim of this study was to evaluate the effect of Floragenist Blossom and one of its main phytochemical naringin on the Corneodesmosin synthesis on keratinocytes.

RESULTS & DISCUSSION

Floragenist Blossom at 0.06% stimulated Corneodesmosin synthesis by 2.3 times (± SEM 25%, p<0.001) compared to the untreated control normalized at 100% (SEM 2%) (Fig. 11).
Additionally, the naringin previously identified in the Floragenist Blossom was tested *in vitro* in the same model.
Naringin at isodose of the Floragenist Blossom, increased Corneodesmosin synthesis by 1.6 times (± SEM 11%, p<0.001) compared to the Untreated control normalized at 100% (SEM 2%) (Fig. 11).

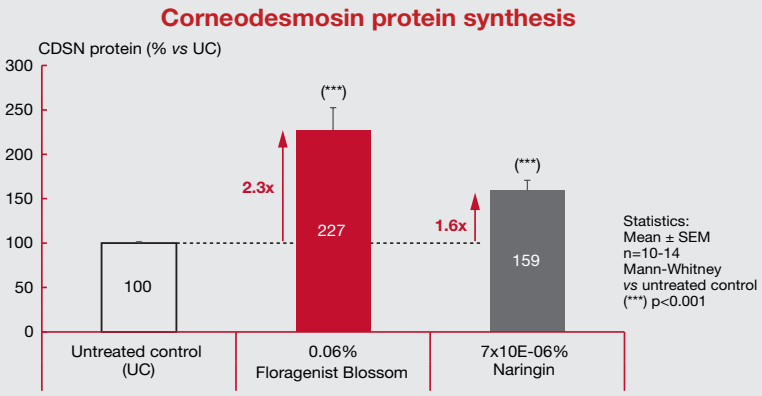


Figure 11 - Quantification of Corneodesmosin synthesis by keratinocytes after treatments with Floragenist Blossom or Naringin.

CONCLUSION

In vitro, Floragenist Blossom stimulated the synthesis of corneodesmosin protein by keratinocytes and may contribute to optimal barrier function. Naringin may supports in part this activity.

MATERIALS&METHODS

Study number

EMB180601-2022-746-CC

Tested products

Coconut-water-*Citrus-aurantium* (batch PG22083.01; INCI: *Cocos Nucifera* (Coconut) Fruit Extract (and) *Citrus Aurantium* Amara (Bitter Orange) Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Blossom at 0.06%. Naringin was tested at 7x10E-06%corresponding to isodose of the Floragenist blossom at 0.06%.

Corneodesmosin protein quantification

Normal human keratinocytes were incubated 3 days at 37°C and 5% CO_2 in MCDB153 medium in the presence of Floragenist Rose or without any product (untreated control). Lack of cytotoxicity of the tested products was checked by MTT test (data not shown). The cellular Corneodesmosin was measured using a specific commercial ELISA method (Sigma).

Results and statistics

Results were normalized to the untreated control and expressed as mean (%) ± standard error of the mean (SEM). Statistical analysis was performed using Mann-Whitney test compared to untreated control. Differences are considered significant for a p-value<0.05.

EFFICACY

Stimulation of lncRNA H19, Desmoglein 1 and Involucrin on keratinocytes

In vitro

OBJECTIVE

LncRNA H19 is the regulator of skin integrity that was previously correlated to keratinocyte differentiation and can promote the increase of *DSG1* expression [Li *et al*, 2017].
So, the aim of this study was to evaluate the effect of Floragenist Blossom on the lncRNA H19 expression and its target *DSG1* on keratinocytes under differentiation process monitored by *IVL* level increase.

RESULTS & DISCUSSION

We visualized the effect of Floragenist Blossom treatment on the lncRNA H19 in keratinocytes using immunohistochemistry Floragenist Blossom increased the lncRNA H19 (red fluorescence) in keratinocyte culture (Fig. 12).
This increase of lncRNA H19 level was quantified by qRT-PCR (Fig. 13). Moreover, we correlated this effect with its *DSG1* mRNA target, and the differentiation state based on *IVL* mRNA stimulation.
Floragenist Blossom increased significantly the expression of the differentiation gene *IVL* (1.6x, p<0.01) as well as the lncRNA H19 (7.5x, p<0.001) and its target *DSG1* (2.4x, p<0.01).

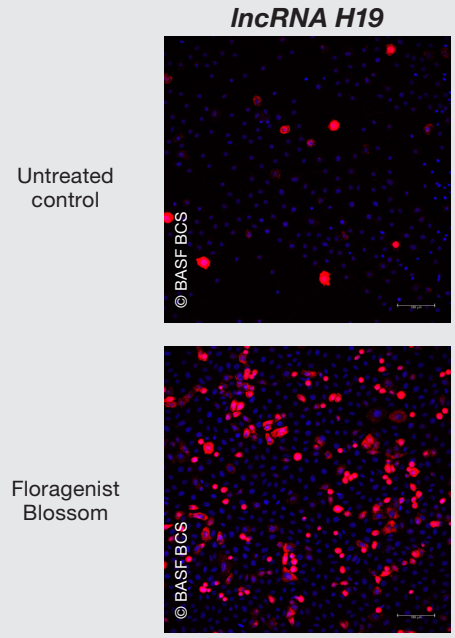


Figure 12 - Immunohistochemical visualization of lncRNA H19 (red fluorescence) in keratinocyte culture before and after Floragenist Blossom treatment. Nuclei were stained with DAPI in blue. Illustrative picture.

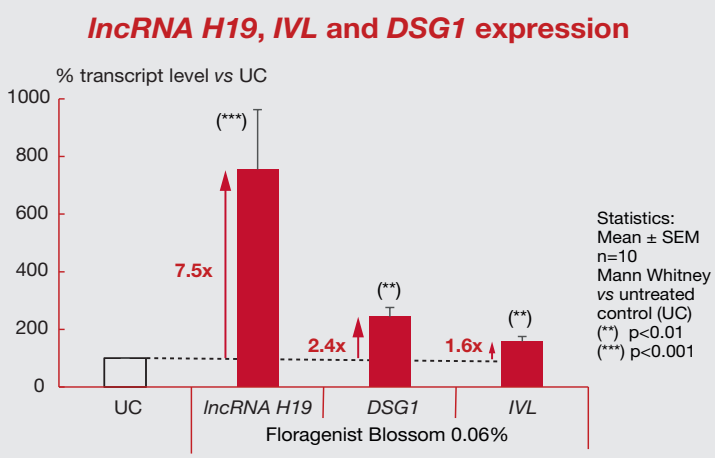


Figure 13 - Floragenist Blossom stimulation of *IVL*, *DSG1* and lncRNA H19 expression by keratinocytes.

CONCLUSION

In keratinocytes, Floragenist Blossom increased lncRNA H19 expression together with the mRNA levels of its reported target *DSG1* and the differentiation marker *IVL*, compared with untreated control in the assay.

MATERIALS&METHODS

Study number
TME2021-008-2024-BM-P02-E03
EMB180601-2022-770-BM 2025

Tested products
Coconut-water-*Citrus-aurantium* (batch PG22083.01; INCI: *Cocos Nucifera* (Coconut) Fruit Extract (and) *Citrus Aurantium* Amara (Bitter Orange) Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Blossom at 0.06%.

Cell culture
Normal human keratinocytes (NHEK), obtained from healthy donors, were seeded in 24-well plates at a rate of 35,000 cells/cm² and cultured in MCDB153 supplemented with 2% FCS, 0.5% antibiotic 96h at 37°C, under 5% CO_2. Then cells were incubated with the products or without any product in MCDB153 supplemented with 0.5% antibiotic.

lncRNA visualization
After human keratinocyte culture and fixation, slides were washed twice in PBS and then dehydrated by successive 5 minutes incubations in ethanol at 50%; 70%; and twice at 100% before drying at room temperature.
In situ hybridization were performed using the RNAscope Plus smRNA-RNA HD Reagent Kit (ACD Biotechne), according to the manual instructions. Shortly, endogenous peroxidase was inactivated by incubation with Hydrogen Peroxide solution for 10 minutes at room temperature and then cells were permeabilized by Protease III treatment for 30 minutes at room temperature before 2 PBS washings. Then, a specific probe against lncRNA H19 was incubated on cells for 2 hours at 40°C before 2 buffer washings.
Signal amplification was performed for each probe using a 3 steps revelation, the last one corresponding to a specific fluorochrome (TSA Vivid Florochrome 650). After washing, cell nuclei were labelled by DAPI and then slides mounted. Fluorescent image captures were performed using a Leica SP8 microscope.

RNA extraction and analysis
Long and short RNAs from keratinocyte cultures were extracted with NucleoSpin miRNA kit (Machery-Nagel), according to the manual instructions. Quantification and quality control of RNAs were performed by measurement of optical densities at 260 and 280 nm.

qRT-PCR for epidermal gene expression
Reverse transcriptions were realized on 0.1 µg of long RNA in 20 µl final mix using SuperScript IV VILO Master Mix Kit (Invitrogen) and the C100 Touch Thermocycleur (BioRad) for each condition.
Quantitative PCRs were performed on 2 µl of RT mix, diluted at 1/2 in 10 µl final reaction, according to manual instruction (LC 480 SYBR Green I Master, Roche) in a LC480 thermocycler (Roche).
Each strain analysis was performed in duplicate and the mean used. Results were normalized compared to the mean of 2 housekeeping genes analyzed in duplicate: *ACTB* and *SDHA* (coding for Actin B and succinate dehydrogenase subunit A, respectively). For each target, the gene expression modulation was calculated using the ΔΔCt method.

Results and statistics
The results are expressed as mean and percentage ± SEM (n=10) compared to untreated control condition (UC) standardized to 100%. The statistical analysis was done using Mann-Whitney Rank Sum Test *vs* UC, with a threshold of significance set at 5% ($p<0.05$).

EFFICACY

Stimulation of 12 genes implicated in cohesion and differentiation process

In vitro

OBJECTIVE

The barrier function of the epidermis involves the expression of numerous tissue-specific genes. To establish and constantly maintain the barrier, keratinocytes undergo a highly controlled and organized differentiation program leading to cornification and desquamation. As differentiation progresses, most of the genes implicated in the epidermal barrier are specifically expressed in the late steps of keratinocytes differentiation [Toulza *et al,* 2007].
So, the aim of this study was to evaluate the effect of Floragenist Blossom on genes related to differentiation and cohesion in keratinocytes, using a thematic Q-RT-PCR assay.

RESULTS & DISCUSSION

Using a thematic Q-RT-PCR assay, we evaluated the level of stimulation of 12 genes related to differentiation and cohesion in keratinocytes by Floragenist Blossom treatment (Fig. 14).
The level of gene expression is expressed in fold-change vs the untreated control (UC) and organized Figure 14 in circular area corresponding to a stimulation scale starting from 2x to 50x.
For example, Floragenist Blossom increased significantly the expression of the differentiation gene (brown) *IVL* (1.7x, p<0.05) presented in the first circle of stimulation (Fig. 14) as well as the gene *LOR* (41.5x, p<0.05) presented in the third circle of stimulation (Fig. 14). Similarly, cohesion genes (green) such as *CLDN1* (1.7x, p<0.05) or *DSC1* (3.8x, p<0.05) are respectively presented in different circles of stimulation (Fig. 14).

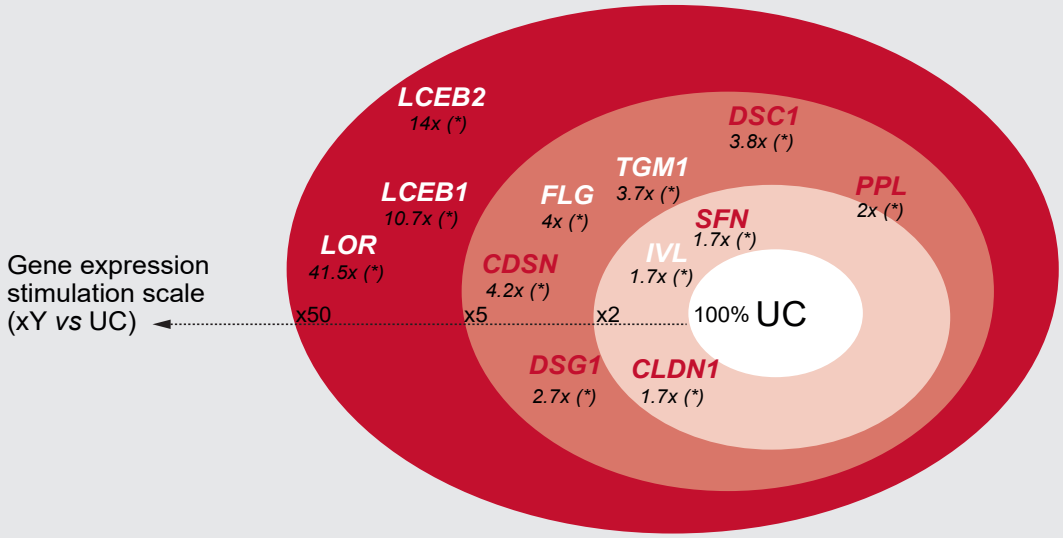


Figure 14 - Gene expression level according to different threshold of stimulation <2x, >2x, >5x and < 50x fold change *vs* UC normalized at 100%.

CONCLUSION

Floragenist blossom upregulates 12 genes involved in differentiation and cohesion of the skin barrier to improve the epidermis organization and support barrier improvement.

MATERIALS&METHODS

Study number

EMB180601-2022-770-BM 2025

Tested products

Coconut-water-*Citrus-Aurantium* (batch JVEGID3194AA / 6GAPEID06900 INCI: *Cocos Nucifera* (Coconut) Fruit Extract (and) *Citrus Aurantium* Amara (Bitter Orange) Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Blossom at 0.06%.

Cell culture

Normal human keratinocytes (NHEK), obtained from healthy donors, were seeded in 24-well plates at a rate of 35,000 cells/cm² and cultured in MCDB153 supplemented with 2% FCS, 0.5% antibiotic 96h at 37°C, under 5% CO_2. Then cells were incubated with the products or without any product in MCDB153 supplemented with 0.5% antibiotic.

RNA extraction and analysis

Total RNAs from keratinocyte cultures were extracted with NucleoSpin miRNA kit (Machery-Nagel), according to the manual instructions. Quantification and quality control of RNAs were performed by measurement of optical densities at 260 and 280 nm.

Thematic qRT-PCR for 12 epidermal gene expression

Reverse transcriptions were realized on 0.1 µg of RNA in 20 µl final mix using SuperScript IV VILO Master Mix Kit (Invitrogen) and the C100 Touch Thermocycleur (BioRad) for each condition.

Quantitative PCRs were performed on 2 µl of RT mix diluted at 1/2 in 10 µl final reaction, according to manual instruction (LC 480 SYBR Green I Master, Roche) in a LC480™ thermocycler (Roche). Dilutions at 1/20 were used for some *FLG*, *LOR*, *LCE1B*, *CDSN* analyses and at 1/200 for some *LCE2B* studies (according to the studied culture).

Each strain analysis was performed in duplicate and the mean used. Results were normalized compared to the mean of 2 housekeeping genes analyzed in duplicate: *ACTB* and *SDHA* (coding for Actin B and succinate dehydrogenase subunit A, respectively). For each target, the gene expression modulation was calculated using the ΔΔCt method.

Results and statistics

The results are expressed as mean and percentage ± SEM (n=4) compared to untreated control condition (UC) standardized to 100%. The statistical analysis was done using Mann-Whitney Rank Sum Test *vs* UC, with a threshold of significance set at 5% ($p<0.05$).

EFFICACY

Barrier strengthening

In vivo

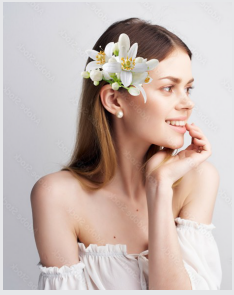
OBJECTIVE

Protecting the skin's barrier integrity is essential to maintain its shield function from external aggressors [Del rosso *et al*, 2016; Krutmann *et al*, 2011]. A healthy barrier also limits trans-epidermal water loss, helping the skin stay hydrated and resilient. The barrier disruption makes the skin more vulnerable to microbial imbalance and external damage [Lee *et al*, 2022; Baker *et al*, 2023; Altemus *et al*, 2001]. Compromised barrier function is linked to conditions such as sensitive skin, xerosis, contact dermatitis, atopic dermatitis, and psoriasis [Del rosso *et al*, 2016; Baker *et al*, 2023].

The clinical study was conducted to evaluate the barrier protection efficacy of Floragenist Blossom at 0.1%, as compared to baseline and placebo (Fig 15).

The study was a double-blind, placebo controlled, split-face, and randomized study of 19 female subjects aged 33-65 years

Placebo



Floragenist Blossom at 0.1%

Self-application was done by volunteers twice a day
Product was applied for four weeks (28 days) before barrier challenge test with SLS washing
Measurement of TEWL prior and immediately after barrier disruption by SLS

Figure 15 - Clinical protocol.

RESULTS & DISCUSSION

After washing with SLS, Floragenist Blossom at 0.1% showed a significant decrease ($p<0.05$) of 52% in TEWL as compared to placebo (Fig. 16), following 28 days of treatment.

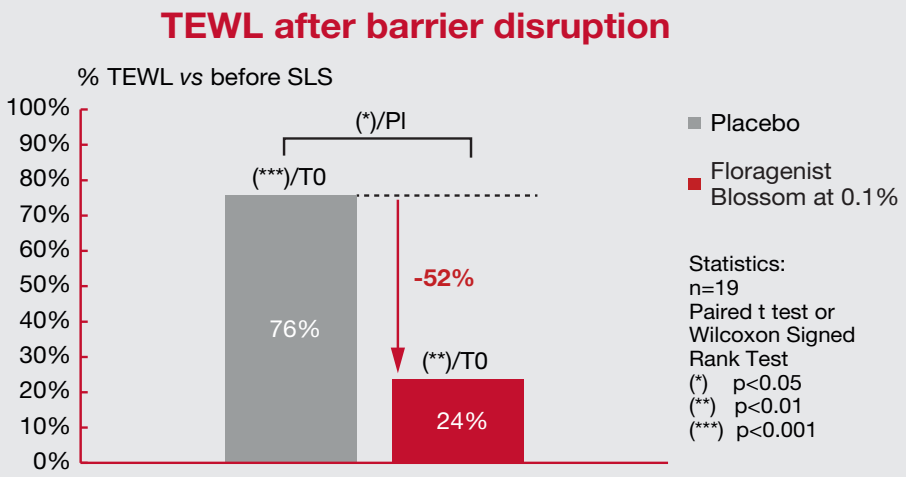


Figure 16 - Change in TEWL after barrier disruption.

CONCLUSION

After 28 days of application, the post-SLS TEWL response was improved for the Floragenist Blossom compared to the placebo, thanks to a reduction of SLS challenge effect on TEWL parameter.

MATERIALS&METHODS

Study number
EMB180601

Tested product
Floragenist Blossom BC10191 at 0.1% (batch C221017.02).

Study design
The clinical study was carried out as a placebo-controlled double-blind randomized split-face study. The barrier protection efficacy of Floragenist Blossom BC10191 at 0.1%, was conducted compared to baseline and placebo.
The study was conducted during a period of 28 days prior barrier challenge test with SLS washing.

Inclusion criteria
35 eligible participants were adults within the age range 18-65. Out of these subjects, 19 met the inclusion criteria at D0 of having a mean TEWL value of left and right sides of the face >=15 and an absolute difference of TEWL value between left and right side of the face ≤ 5 at D0. They agreed to follow strict protocols regarding skincare, product usage, and lifestyle, as detailed in the study.

Application modality
Participants were instructed to apply the product themselves twice daily, once in the morning and once in the evening, for a total duration of four weeks (28 days).

SLS stress application
Following product treatment of 28 days, measurements were taken before and then five minutes after the face was cleansed with a 12% SLS solution. The lower the TEWL, the better the Skin Barrier property.

Aquaflux TEWL probe
The Aquaflux is an instrument which measures the flux density of water via the closed chamber method. Water evaporates into the measurement chamber, until it meets the condenser at the other end of the measurement chamber and is turned to ice. Sensors within the walls of the chamber measure the flux density, relative humidity and temperature of the chamber. Because the Aquaflux the measures flux density, it can be used to measure the Trans-Epidermal Water Loss (TEWL) of the skin. The rate of water loss across the epidermal layer of the skin is an important indicator of skin barrier function.
The lower the TEWL, the better the Skin Barrier property
The Aquaflux was used to assess the changes in skin barrier function before and after barrier disruption with SLS solution. Each cheek was measured twice.

Results and statistics
The %TEWL calculation was performed using the following equation:
%TEWL = [Average(Tx-T0)*100/AverageT0]%
T0: measurement values after 28 days of treatment (before barrier disruption by SLS washing).
Tx: measurement values at 5 mins after barrier disruption from SLS solution.

All data has been analyzed to confirm the normality of the distribution using the Shapiro-Wilk Test.

The statistical analysis of the change in the studied parameters for each product over time as well as differences in the studied parameters between the treatment groups were performed as follows:
1) Validation of the normality of the studied parameters: Paired t test.
2) Invalidation of the normality of the parameters studied: Wilcoxon Signed Rank Test. All statistical tests were run at 95% confidence level (p-value <0.05).



GENERAL CONCLUSION

Floragenist Blossom: the Skin Guardian Elixir

A new-generation active that helps delicate skin prepare itself to face daily challenges. thanks to its “Adaptive Skin Strength” improving TEWL parameter.

Sourced from Morocco and re-engineered with BASF’s proprietary AquaGenesis extraction technology, Floragenist Blossom reinvents an iconic flower into a boosted, high performance elixir delivering its full potential to preserve skin integrity.

Today’s consumers are increasingly looking for botanicals that feel familiar and authentic but also come with transparent sourcing and robust scientific evidence. Floragenist Blossom was developed to meet this expectation with a traceable orange blossom extract and clinically relevant barrier readouts.

Fragile or sensitive skins present difficulties to face daily stresses thanks to their adaptive system decline that results in an increased permeability.

Floragenist Blossom is a next-generation of natural flower extract sourced from Morocco, combining sustainable and ethical floral culture. AquaGenesis extraction technology, based on natural solvent, is employed to enhance the concentration in Flavanone glycosides to offer perfect shield preserving barrier integrity for fragile skins.

In vitro, assessments showed that Floragenist Blossom supports skin barrier robustness across multiple stress paradigms, including SLS induced cleansing disruption, recovery of key barrier proteins after mechanical stripping, and maintenance of Filaggrin levels under bacterial disturbance. More specifically, the data indicate stimulation of skin integrity proteins (including Involucrin, Filaggrin, Desmoglein 1 and Corneodesmosin), together with an epigenetic contribution via lncRNA H19, supporting epidermal cohesion and adaptive skin strength. Additionally, Floragenist Blossom upregulated a panel of genes involved in epidermal differentiation and cohesion (e.g., Loricrin and Claudin 1), which are essential for maintaining epithelial integrity.

Tested *in vivo* for 28 days in a double-blind, placebo controlled, split-face and randomized study of 19 females, Floragenist Blossom at 0.1% - applied prior to chemical stress - significantly improves skin barrier protection by decreasing by 52% the TEWL parameter vs placebo, making it an effective pre-treatment to support the skin’s adaptive response under stress.

Designed for fragile or reactive skin, Floragenist Blossom combines AquaGenesis coconut water extraction with epigenetic precision (lncRNA H19) to help reinforce epidermal cohesion and improve barrier resilience.

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and regulatory product information
- Toxicological abstract
- Composition data sheet
- Specifications
- Product Purity Profile
- Formulation Data Sheet
- Natural content (according ISO 16128)
- Vegan friendly statement

Annex 2 - Clinical test formula

Trade name	INCI name	Function	Placebo %	Floragenist Blossom %
Water	Water	Solvent	75.24	75.14
Eumulgin SG	Sodium Stearoyl Glutamate	Emulsifier	0.40	0.40
Glycerin 99-5	Glycerin	Humectant	2.00	2.00
Euxyl K 903 (Ashland)	Benzyl Alcohol, Benzoic Acid, Dehydroacetic Acid, Tocopherol	Preservative	1.20	1.20
Emulgade PL 68/50	Cetearyl Glucoside, Cetearyl Alcohol	Emulsifier	2.00	2.00
Cutina HVG	Cetyl PEG/PPG-10/1 Dimethicone	Consistency agent	2.00	2.00
Cegesoft PS 6	Olus Oil [EU], Vegetable Oil [CTFA]	Emollient	3.00	3.00
Cetiol 4 All	Dipropylheptyl Carbonate	Emollient	3.00	3.00
Cetiol Sensoft	Propylheptyl Caprate	Emollient	5.00	5.00
Cosmedia SP	Sodium Polyacrylate	Rheology modifier	1.00	1.00
Cetiol iSan	Tridecane	Emollient	4.00	4.00
Floragenist Blosssom BC10191	-	Active ingredient	0.00	0.10
Citric Acid Solution 10%	Citric Acid, Water	pH adjustement	1.16	1.16

Annex 3 - Formulation example

Adaptive dew cream (SC-FR-26-NK-0174-003)

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	68.65	Solvent
A	1,3-Butanediol	Butylene Glycol	5.00	Humectant
A	euxyl PE 9010	Phenoxyethanol, Ethylhexylglycerin	1.00	Preservative
A	Glycerin (99.5%)	Glycerin	5.00	Humectant
A	Floragenist™ Blossom BC10191	Maltodextrin, Cocos Nucifera (Coconut) Fruit Extract, Citrus Aurantium Amara (Bitter Orange) Flower Extract	0.10	Active Ingredient
B	Eumulgin® SG	Sodium Stearoyl Glutamate	0.30	Emulsifier
B	Cegesoft® PS 6	Olus (Vegetable) Oil	3.00	Emollient
B	Cetiol® C 5C	Coco-Caprylate/Caprate	5.00	Emollient
B	Verdessence® Alginate	Algin	0.60	Rheology Modifier
B	Verdessence® Xanthan	Xanthan Gum	0.60	Rheology Modifier
C	Water, demin.	Aqua	10.00	Solvent
C	Calcium Chloride Dihydrate	Calcium Chloride	0.10	Stabilizer
D	Perfume "Fleur d'oranger"	Parfum	0.20	Perfume / Fragrance
E	Citric Acid (10% solution)	Citric Acid	0.45	pH Adjustment

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Moon reset balm (SC-FR-26-NK-0174-002)

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	71.50	Solvent
A	Eumulgin® SG	Sodium Stearoyl Glutamate	0.40	Emulsifier
A	Glycerin (99.5%)	Glycerin	3.00	Humectant
A	Floragenist™ Blossom BC10191	Maltodextrin, Cocos Nucifera (Coconut) Fruit Extract, Citrus Aurantium Amara (Bitter Orange) Flower Extract	0.10	Active Ingredient
A	euxyl PE 9010	Phenoxyethanol, Ethylhexylglycerin	1.00	Preservative
A	Verdessence® Xanthan	Xanthan Gum	0.20	Rheology Modifier
B	Emulgade® PL 68/50	Cetearyl Glucoside, Cetearyl Alcohol	2.00	Emulsifier
B	Cutina® HVG	Hydrogenated Vegetable Glycerides	2.00	Consistency Agent
B	Cetiol® Sensoft	Propylheptyl Caprylate	5.00	Emollient
B	Cosmedia® SP	Sodium Polyacrylate	0.60	Rheology Modifier
B	Cetiol® LC	Coco-Caprylate/Caprate	5.00	Emollient
B	Monomuls® 90-O 18	Glyceryl Oleate	1.00	Emollient
B	Cegesoft® SB 45 TR	Butyrospermum Parkii (Shea) Butter	2.00	Emollient
B	Cutina® PES	Pentaerythrityl Distearate	2.00	Consistency Agent
C	Cetiol® iSan	Tridecane	4.00	Emollient

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